October 26, 2026

PoODLE

Developed By- Michigan Department of Health & Human Services

Advanced Molecular Detection Southeast Region

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Advanced Molecular Detection
Southeast Region Bioinformatics

OVERVIEW

Purpose:

POODLE, developed by the Michigan Department of Health & Human Services, provides high-throughput bacterial SNP-based analysis, enabling variant calling, recombination filtering, phylogenetics, linkage evaluation, pangenome assessment, and integrated reporting for genomic surveillance and outbreak investigations.

Usage:

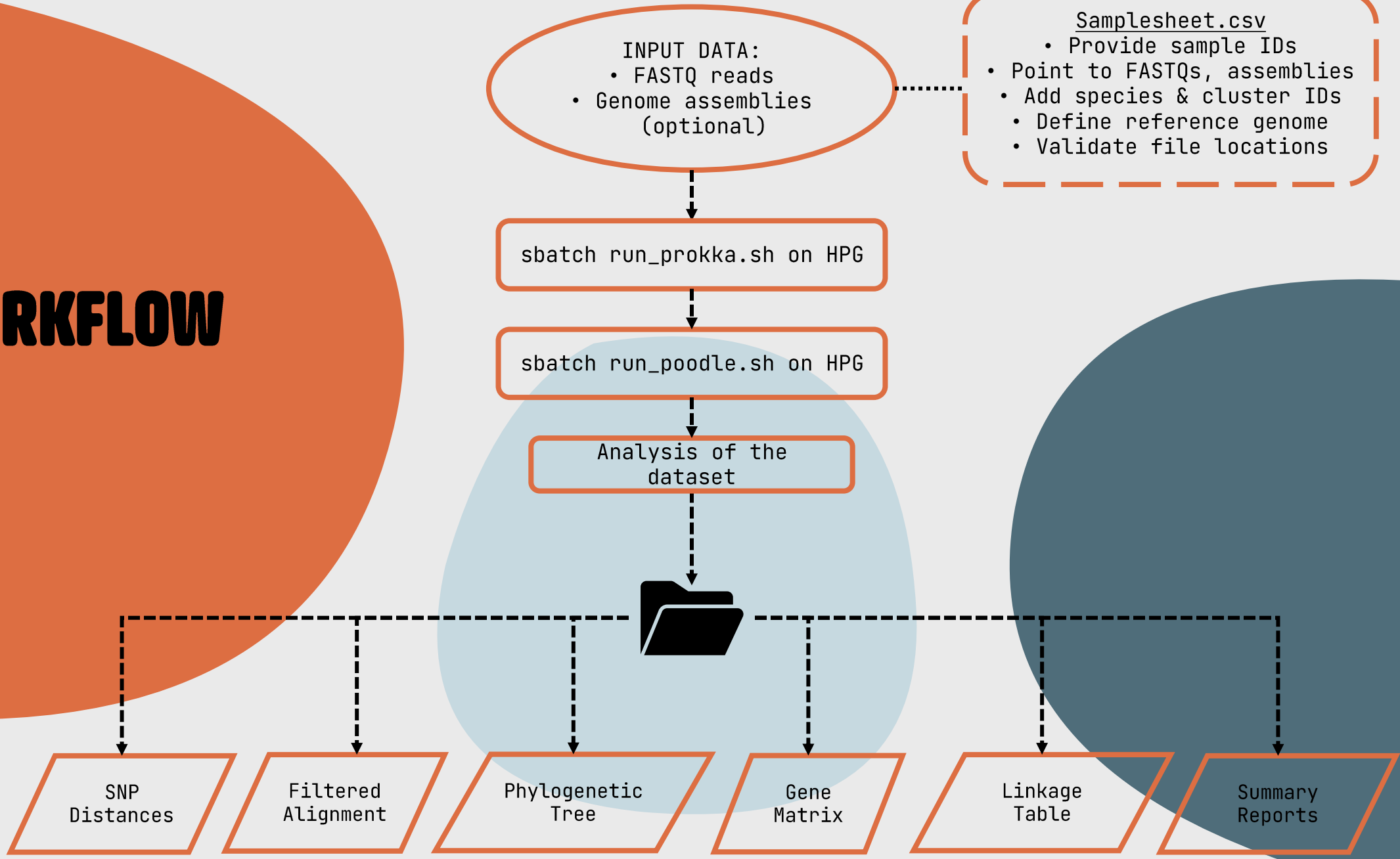
Performs read QC, reference-based alignment, core-SNP extraction, SNP distance analysis, recombination-aware tree building, pangenome reconstruction, and automated YAML/Quarto reporting to support routine public-health genomic workflows.

Dependencies:

Nextflow
Conda
Apptainer/Singularity

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WORKFLOW



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APPLICATION

Objective

Use various paired FASTQ reads (Illumina) and analyze the dataset using Poodle pipeline.

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APPLICATION CONT.

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/
git clone https://github.com/MDHHS-Bioinformatics/poodle
mkdir analysis/
cd analysis/
cp /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

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assets		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
bin		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
conf		9/2/2026 1:15:36 PM	rwxr-sr-x	n.yengalareddy
docs		9/2/2026 1:27:14 PM	rwxr-sr-x	n.yengalareddy
input		9/1/2026 2:28:15 PM	rwxr-sr-x	n.yengalareddy
lib		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
modules		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
output		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
quarto_templates		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
work		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
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run_poodle.sh	2 KB	9/1/2026 2:38:36 PM	rw-r--r--	n.yengalareddy
run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/

Name	Size	Changed	Rights	Owner
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annotations		9/1/2026 2:54:40 PM	rwxr-sr-x	n.yengalareddy
fastq		9/1/2026 8:36:10 AM	rwxr-sr-x	n.yengalareddy
samplesheet.csv	7 KB	9/2/2026 9:04:21 AM	rw-r--r--	n.yengalareddy

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
annotations		9/1/2026 2:54:40 PM	rwxr-sr-x	n.yengalareddy
fastq		9/1/2026 8:36:10 AM	rwxr-sr-x	n.yengalareddy
samplesheet.csv	7 KB	9/2/2026 9:04:21 AM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/annotations/

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SRR40019074		9/1/2026 2:52:54 PM	rwxr-sr-x	n.yengalareddy
SRR40019075		9/1/2026 2:53:47 PM	rwxr-sr-x	n.yengalareddy
SRR40019078		9/1/2026 2:54:40 PM	rwxr-sr-x	n.yengalareddy
SRR40019080		9/1/2026 2:55:31 PM	rwxr-sr-x	n.yengalareddy
SRR40064142		9/1/2026 2:48:02 PM	rwxr-sr-x	n.yengalareddy
SRR40064153		9/1/2026 2:48:45 PM	rwxr-sr-x	n.yengalareddy
SRR40064187		9/1/2026 2:49:31 PM	rwxr-sr-x	n.yengalareddy
SRR40064547		9/1/2026 2:50:18 PM	rwxr-sr-x	n.yengalareddy
SRR40064574		9/1/2026 2:51:08 PM	rwxr-sr-x	n.yengalareddy
prokka_samplesheet.csv	2 KB	9/1/2026 2:28:15 PM	rw-r--r--	n.yengalareddy

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SRR40064153     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/assemblies/SRR40064153.fna      Acinetobacter   baumannii
SRR40064187     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/assemblies/SRR40064187.fna      Acinetobacter   baumannii
SRR40064547     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/assemblies/SRR40064547.fna      Acinetobacter   baumannii
SRR40064574     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/assemblies/SRR40064574.fna      Acinetobacter   baumannii
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SRR40019075     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019075.fna      Shigella        sonnei
SRR40019078     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019078.fna      Shigella        sonnei
SRR40019080     /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019080.fna      Shigella        sonnei
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
annotations		9/1/2026 2:54:40 PM	rwxr-sr-x	n.yengalareddy
fastq		9/1/2026 8:36:10 AM	rwxr-sr-x	n.yengalareddy
samplesheet.csv	7 KB	9/2/2026 9:04:21 AM	rw-r--r--	n.yengalareddy

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SRR40019073_R2.fastq.gz	229,033 KB	9/1/2026 8:36:04 AM	rw-r--r--	n.yengalareddy
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SRR40019074_R2.fastq.gz	193,298 KB	9/1/2026 8:36:06 AM	rw-r--r--	n.yengalareddy
SRR40019075_R1.fastq.gz	148,617 KB	9/1/2026 8:36:06 AM	rw-r--r--	n.yengalareddy
SRR40019075_R2.fastq.gz	166,056 KB	9/1/2026 8:36:07 AM	rw-r--r--	n.yengalareddy
SRR40019078_R1.fastq.gz	163,000 KB	9/1/2026 8:36:08 AM	rw-r--r--	n.yengalareddy
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SRR40019080_R1.fastq.gz	175,631 KB	9/1/2026 8:36:10 AM	rw-r--r--	n.yengalareddy
SRR40019080_R2.fastq.gz	207,906 KB	9/1/2026 8:36:11 AM	rw-r--r--	n.yengalareddy
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SRR40064153_R2.fastq.gz	299,955 KB	9/1/2026 8:35:55 AM	rw-r--r--	n.yengalareddy
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SRR40064547_R1.fastq.gz	322,026 KB	9/1/2026 8:35:57 AM	rw-r--r--	n.yengalareddy
SRR40064547_R2.fastq.gz	327,354 KB	9/1/2026 8:35:58 AM	rw-r--r--	n.yengalareddy
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florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019075.fna,HC42-C1,Shigella_sonnei,/blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019075.fna
SRR40019078,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/fastq/SRR40019078_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/poodle1026/input/fastq/SRR40019078_R2.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/annotations/SRR40019078/SRR40019078.gff,/blue/bphl-
florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019078.fna,HC42-C1,Shigella_sonnei,/blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019075.fna
SRR40019080,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/fastq/SRR40019080_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/poodle1026/input/fastq/SRR40019080_R2.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/annotations/SRR40019080/SRR40019080.gff,/blue/bphl-
florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019080.fna,HC42-C1,Shigella_sonnei,/blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/corge0928/output-20260812164858/Shigella_sonnei/assemblies/SRR40019075.fna
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

Name	Size	Changed	Rights	Owner
..		9/4/2026 11:16:24 AM	rwxr-sr-x	n.yengalareddy
assets		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
bin		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
conf		9/2/2026 1:15:36 PM	rwxr-sr-x	n.yengalareddy
docs		9/2/2026 1:27:14 PM	rwxr-sr-x	n.yengalareddy
input		9/1/2026 2:28:15 PM	rwxr-sr-x	n.yengalareddy
lib		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
modules		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
output		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
quarto_templates		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
work		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
workflows		9/2/2026 1:19:55 PM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATIONS.md	4 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	2 KB	9/2/2026 1:19:11 PM	rw-r--r--	n.yengalareddy
modules.json	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
nextflow.config	7 KB	9/2/2026 1:16:07 PM	rwxr-xr-x	n.yengalareddy
nextflow_schema.json	17 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
poodle.40887634.err	1 KB	9/2/2026 10:50:11 AM	rw-r--r--	n.yengalareddy
poodle.40887634.out	91 KB	9/2/2026 10:58:00 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
README.md	11 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
run_poodle.sh	2 KB	9/1/2026 2:38:36 PM	rw-r--r--	n.yengalareddy
run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/conf/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
base.config	3 KB	9/2/2026 1:15:36 PM	rw-r--r--	n.yengalareddy
modules.config	8 KB	8/31/2026 12:58:43 PM	rwxr-xr-x	n.yengalareddy
test.config	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
test_full.config	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/report/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
quarto_both.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_gub_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_mash_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_neither.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
yaml_both.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_gub_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_mash_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_neither.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/iqtree/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
tests		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
environment.yml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	3 KB	9/2/2026 1:21:14 PM	rw-r--r--	n.yengalareddy
meta.yml	12 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/gubbins/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
tests		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
environment.yml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	3 KB	9/2/2026 10:29:37 AM	rw-r--r--	n.yengalareddy
meta.yml	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/subworkflows/local/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
input_check.nf	6 KB	8/31/2026 12:58:43 PM	rwxr-xr-x	n.yengalareddy
snippyclusters.nf	8 KB	9/1/2026 1:52:17 PM	rwxr-xr-x	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/workflows/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
poodle.nf	15 KB	9/2/2026 1:19:55 PM	rwxr-xr-x	n.yengalareddy

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/conf/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
base.config	3 KB	9/2/2026 1:15:36 PM	rw-r--r--	n.yengalareddy
modules.config	8 KB	8/31/2026 12:58:43 PM	rwxr-xr-x	n.yengalareddy
test.config	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
test_full.config	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

```
/*
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    MDHHS-Bioinformatics/poodle Nextflow base config file
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    A 'blank slate' config file, appropriate for general use on most high performance
    compute environments. Assumes that all software is installed and available on
    the PATH. Runs in `local` mode - all jobs will be run on the logged in environment.
----------------------------------------------------------------------------------------
*/

process {

    // Defaults for all processes
    cpus   = { 1    * task.attempt }
    memory = { 6.GB * task.attempt }
    time   = { 4.h  * task.attempt }

    errorStrategy = { task.exitStatus in [143,137,104,134,139] ? 'retry' : 'finish' }
    maxRetries    = 1
    maxErrors     = '-1'

    // Process-specific resource requirements
    // NOTE - Please try and re-use the labels below as much as possible.
    //        These labels are used and recognised by default in DSL2 files hosted on nf-core/modules.
    //        If possible, it would be nice to keep the same label naming convention when
    //        adding in your local modules too.
    // See https://www.nextflow.io/docs/latest/config.html#config-process-selectors
    withLabel:process_single {
        cpus   = { 1 }
        memory = { 6.GB * task.attempt }
        time   = { 4.h  * task.attempt }
    }
    withLabel:process_low {
        cpus   = { 2     * task.attempt }
        memory = { 12.GB * task.attempt }
        time   = { 4.h   * task.attempt }
    }
    withLabel:process_medium {
        cpus   = { 6     * task.attempt }
        memory = { 36.GB * task.attempt }
        time   = { 8.h   * task.attempt }
    }
    withLabel:process_high {
        cpus   = { 12    * task.attempt }
        memory = { 72.GB * task.attempt }
        time   = { 16.h  * task.attempt }
    }
    withLabel:process_long {
        time   = { 20.h  * task.attempt }
    }
    withLabel:process_high_memory {
        memory = { 200.GB * task.attempt }
    }
    withLabel:error_ignore {
        errorStrategy = 'ignore'
    }
    withLabel:error_retry {
        errorStrategy = 'retry'
        maxRetries    = 2
    }
    withName:CUSTOM_DUMPSOFTWAREVERSIONS {
        cache = false
    }
}
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/report/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
quarto_both.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_gub_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_mash_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_neither.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
yaml_both.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_gub_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_mash_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_neither.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

```
process QUARTO_BOTH {
  tag "${meta.species}_${meta.cluster_id}"
  label 'process_medium'

  errorStrategy 'ignore'

  container 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting@sha256:2a3c9d9a87796ff612cce94e7638d906a04aec850ca0358446a3d5415526b326'
  // 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting:1.0.0'

  containerOptions(
    ['singularity', 'apptainer'].contains(workflow.containerEngine)
      ? '--no-home --env QUARTO_PRINT_STACK=true'
      : '--user $(id -u):$(id -g) -e QUARTO_PRINT_STACK=true'
  )

  stageInMode 'copy'
  afterScript 'rm -rf tmp'

  input:
  tuple val(meta), path(linkages), path(snptree), path(snpmatrix),
        path(pan_summary), path(pan_roary), path(pan_rtab), path(pan_genedists),
        path(gubtree), path(gubmatrix),
        path(mashtree), path(mashmatrix)
  path(yaml)
  path(qmd)
  path(logo)

  output:
  tuple val(meta), path("${meta.species}_${meta.cluster_id}.html"), emit: html
  path "versions.yml"                                              , emit: versions

  script:
  def prefix = task.ext.prefix ?: "${meta.species}_${meta.cluster_id}"
  cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
  species = task.ext.prefix ?: "${meta.species}"
  """
  mkdir -p .quarto-cache .quarto-data .deno-cache tmp
  chmod -R u+rwX .quarto-cache .quarto-data .deno-cache tmp

  export XDG_CACHE_HOME="\$PWD/.quarto-cache"
  export XDG_DATA_HOME="\$PWD/.quarto-data"
  export XDG_RUNTIME_DIR="\$PWD/tmp"
  export DENO_DIR="\$PWD/.deno-cache"
  export QUARTO_PRINT_STACK=true

  cp -n ${linkages} ${snptree} ${snpmatrix} ${pan_summary} ${pan_roary} ${pan_rtab} ${pan_genedists} \\
        ${gubtree} ${gubmatrix} ${mashtree} ${mashmatrix} ${logo} .

  quarto render $qmd --execute-params $yaml --output "${prefix}.html"

  cat <<-END_VERSIONS > versions.yml
  "${task.process}":
      quarto: \$(quarto --version)
  END_VERSIONS
  """
}
```

```
process QUARTO_GUB_ONLY {
  tag "${meta.species}_${meta.cluster_id}"
  label 'process_medium'

  errorStrategy 'ignore'

  container 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting@sha256:2a3c9d9a87796ff612cce94e7638d906a04aec850ca0358446a3d5415526b326'
  // 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting:1.0.0'

  containerOptions(
    ['singularity', 'apptainer'].contains(workflow.containerEngine)
      ? '--no-home --env QUARTO_PRINT_STACK=true'
      : '--user $(id -u):$(id -g) -e QUARTO_PRINT_STACK=true'
  )

  stageInMode 'copy'
  afterScript 'rm -rf tmp'

  input:
  tuple val(meta), path(linkages), path(snptree), path(snpmatrix),
        path(pan_summary), path(pan_roary), path(pan_rtab), path(pan_genedists),
        path(gubtree), path(gubmatrix)
  path(yaml)
  path(qmd)
  path(logo)

  output:
  tuple val(meta), path("${meta.species}_${meta.cluster_id}.html"), emit: html
  path "versions.yml", emit: versions

  script:
  def prefix = "${meta.species}_${meta.cluster_id}"
  cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
  species = task.ext.prefix ?: "${meta.species}"
  """
  mkdir -p .quarto-cache .quarto-data .deno-cache tmp
  chmod -R u+rwX .quarto-cache .quarto-data .deno-cache tmp

  export XDG_CACHE_HOME="\$PWD/.quarto-cache"
  export XDG_DATA_HOME="\$PWD/.quarto-data"
  export XDG_RUNTIME_DIR="\$PWD/tmp"
  export DENO_DIR="\$PWD/.deno-cache"
  export QUARTO_PRINT_STACK=true

  cp -n ${linkages} ${snptree} ${snpmatrix} ${pan_summary} ${pan_roary} ${pan_rtab} ${pan_genedists} \\
        ${gubtree} ${gubmatrix} ${logo} .
  quarto render $qmd --execute-params $yaml --output "${prefix}.html"
  cat <<-END_VERSIONS > versions.yml
  "${task.process}":
      quarto: \$(quarto --version)
  END_VERSIONS
  """
}
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/report/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
quarto_both.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_gub_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_mash_only.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
quarto_neither.nf	2 KB	9/2/2026 1:20:33 PM	rw-r--r--	n.yengalareddy
yaml_both.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_gub_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_mash_only.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
yaml_neither.nf	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

```
process QUARTO_MASH_ONLY {
  tag "${meta.species}_${meta.cluster_id}"
  label 'process_medium'

  errorStrategy 'ignore'

  container 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting@sha256:2a3c9d9a87796ff612cce94e7638d906a04aec850ca0358446a3d5415526b326'
  // 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting:1.0.0'

  containerOptions(
    ['singularity', 'apptainer'].contains(workflow.containerEngine)
      ? '--no-home --env QUARTO_PRINT_STACK=true'
      : '--user $(id -u):$(id -g) -e QUARTO_PRINT_STACK=true'
  )

  stageInMode 'copy'
  afterScript 'rm -rf tmp'

  input:
  tuple val(meta), path(linkages), path(snptree), path(snpmatrix),
        path(pan_summary), path(pan_roary), path(pan_rtab), path(pan_genedists),
        path(mashtree), path(mashmatrix)
  path(yaml)
  path(qmd)
  path(logo)

  output:
  tuple val(meta), path("${meta.species}_${meta.cluster_id}.html"), emit: html
  path "versions.yml", emit: versions

  script:
  def prefix = "${meta.species}_${meta.cluster_id}"
  cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
  species = task.ext.prefix ?: "${meta.species}"
  """
  mkdir -p .quarto-cache .quarto-data .deno-cache tmp
  chmod -R u+rwX .quarto-cache .quarto-data .deno-cache tmp

  export XDG_CACHE_HOME="\$PWD/.quarto-cache"
  export XDG_DATA_HOME="\$PWD/.quarto-data"
  export XDG_RUNTIME_DIR="\$PWD/tmp"
  export DENO_DIR="\$PWD/.deno-cache"
  export QUARTO_PRINT_STACK=true

  cp -n ${linkages} ${snptree} ${snpmatrix} ${pan_summary} ${pan_roary} ${pan_rtab} ${pan_genedists} \\
        ${mashtree} ${mashmatrix} ${logo} .

  quarto render $qmd --execute-params $yaml --output "${prefix}.html"

  cat <<-END_VERSIONS > versions.yml
  "${task.process}":
      quarto: \$(quarto --version)
  END_VERSIONS
  """
}
```

```
process QUARTO_NEITHER {
  tag "${meta.species}_${meta.cluster_id}"
  label 'process_medium'

  errorStrategy 'ignore'

  container 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting@sha256:2a3c9d9a87796ff612cce94e7638d906a04aec850ca0358446a3d5415526b326'
  // 'quay.io/mdhhs_bioinformatics/quarto-wgs-reporting:1.0.0'

  containerOptions(
    ['singularity', 'apptainer'].contains(workflow.containerEngine)
      ? '--no-home --env QUARTO_PRINT_STACK=true'
      : '--user $(id -u):$(id -g) -e QUARTO_PRINT_STACK=true'
  )

  stageInMode 'copy'
  afterScript 'rm -rf tmp'

  input:
  tuple val(meta), path(linkages), path(snptree), path(snpmatrix),
        path(pan_summary), path(pan_roary), path(pan_rtab), path(pan_genedists)
  path(yaml)
  path(qmd)
  path(logo)

  output:
  tuple val(meta), path("${meta.species}_${meta.cluster_id}.html"), emit: html
  path "versions.yml", emit: versions

  script:
  def prefix = "${meta.species}_${meta.cluster_id}"
  cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
  species = task.ext.prefix ?: "${meta.species}"
  """
  mkdir -p .quarto-cache .quarto-data .deno-cache tmp
  chmod -R u+rwX .quarto-cache .quarto-data .deno-cache tmp

  export XDG_CACHE_HOME="\$PWD/.quarto-cache"
  export XDG_DATA_HOME="\$PWD/.quarto-data"
  export XDG_RUNTIME_DIR="\$PWD/tmp"
  export DENO_DIR="\$PWD/.deno-cache"
  export QUARTO_PRINT_STACK=true

  cp -n ${linkages} ${snptree} ${snpmatrix} ${pan_summary} ${pan_roary} ${pan_rtab} ${pan_genedists} ${logo} .

  quarto render $qmd --execute-params $yaml --output "${prefix}.html"

  cat <<-END_VERSIONS > versions.yml
  "${task.process}":
      quarto: \$(quarto --version)
  END_VERSIONS
  """
}
```

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Application Cont.

```
process IQTREE {
    tag "${meta.species}_${meta.cluster_id}"
    label 'process_medium'

    conda "${moduleDir}/environment.yml"
    container 'quay.io/biocontainers/iqtree@sha256:604552032e25a7a8d30c8d2f6cbc72b576f2f8159b4d5a0bc17c28dfd9e55511'
    // 'quay.io/biocontainers/iqtree:2.4.0--h503566f_0'

    input:
    tuple val(meta), path(alignment), val(constant_sites)

    output:
    tuple val(meta), path("*.nwk")      , emit: phylogeny     , optional: true
    tuple val(meta), path("*.iqtree")       , emit: report        , optional: true
    path "versions.yml"                     , emit: versions

    when:
    task.ext.when == null || task.ext.when

    script:
    def args                    = task.ext.args           ?: ''
    def alignment_arg           = alignment               ? "-s $alignment": ''
    def args_extension = task.ext.args_extension ?: ''
    prefix = task.ext.prefix ?: "${meta.species}_${meta.cluster_id}"
    cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
    species = task.ext.prefix ?: "${meta.species}"
    def memory                      = task.memory.toString().replaceAll(' ', '')
    """
    # Count number of samples (FASTA format)
    count=\$(grep -c "^>" "$alignment")

    # Perform bootstrapping if there are more than 4 samples
    if [[ \$count -gt 4 ]]; then
        bs="-B 1000"
    else
        bs=""
    fi

    iqtree2 \\
        $args \\
        $alignment_arg \\
        -fconst $constant_sites \\
        -pre ${prefix}${args_extension} \\
        -nt AUTO \\
        -safe \\
        -redo \\
        -m GTR+G4 \\
        -ntmax $task.cpus \\
        -mem $memory \\
        \$bs \\
        -seed 12345

    # rename file for consistency
    if [[ !{count}  < 5 ]]
    then
        mv *.treefile ${prefix}${args_extension}.nwk
    else
        mv *.contree ${prefix}${args_extension}.nwk
    fi

    cat <<-END_VERSIONS > versions.yml
    "${task.process}":
        iqtree: \$(echo \$(iqtree -version 2>&1) | sed 's/^IQ-TREE multicore version //;s/ .*//')
    END_VERSIONS
    """

    stub:
    prefix = task.ext.prefix ?: meta.id
    """
    touch "${prefix}.treefile"
    touch "${prefix}.iqtree"

    cat <<-END_VERSIONS > versions.yml
    "${task.process}":
        iqtree: \$(echo \$(iqtree -version 2>&1) | sed 's/^IQ-TREE multicore version //;s/ .*//')
    END_VERSIONS
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/iqtree/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
tests		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
environment.yml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	3 KB	9/2/2026 1:21:14 PM	rw-r--r--	n.yengalareddy
meta.yml	12 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

Advanced Molecular Detection
Southeast Region Bioinformatics

APPLICATION CONT.

```
process GUBBINS {
    tag "${meta.species}_${meta.cluster_id}"
    label 'process_medium'

    errorStrategy 'ignore'

    conda "${moduleDir}/environment.yml"
    container 'quay.io/biocontainers/gubbins@sha256:8e36a93ce43f63fe466617addd0490641edee0c3030c511166d8171622c8c90c'
    // quay.io/biocontainers/gubbins:3.4.3--py310hfc0ef84_1

    input:
    tuple val(meta), path(msa), val(constant_sites)

    output:
    tuple val(meta), path("*.fasta")                          , emit: fasta
    tuple val(meta), path("*.gff")                            , emit: gff
    tuple val(meta), path("*.vcf")                            , emit: vcf
    tuple val(meta), path("*.csv")                            , emit: stats
    tuple val(meta), path("*.phylip")                         , emit: phylip
    tuple val(meta), path("*.recombination_predictions.embl") , emit: embl_predicted
    tuple val(meta), path("*.branch_base_reconstruction.embl") , emit: embl_branch
    tuple val(meta), path("*.final_tree.tre")                 , emit: tree
    tuple val(meta), path("*.node_labelled.final_tree.tre")   , emit: tree_labelled
    tuple val(meta), path("*.final_bootstrapped_tree.tre")    , emit: bootstrap_tree, optional: true
    path "versions.yml"                                       , emit: versions


    when:
    task.ext.when == null || task.ext.when

    script:
    def args = task.ext.args ?: ''
    prefix = task.ext.prefix ?: "${meta.species}_${meta.cluster_id}"
    cluster_id = task.ext.prefix ?: "${meta.cluster_id}"
    species = task.ext.prefix ?: "${meta.species}"
    """
    mkdir numba_cache_dir
    export NUMBA_CACHE_DIR='./numba_cache_dir'

    run_gubbins.py \\
        --threads $task.cpus \\
        --prefix $prefix \\
        --first-tree-builder iqtree-fast \\
        --tree-builder iqtree \\
        --tree-args " -fconst $constant_sites" \\
        $args \\
        $msa
    cat <<-END_VERSIONS > versions.yml
    "${task.process}":
        gubbins: \$(run_gubbins.py --version 2>&1 | tail -n 1)
    END_VERSIONS
    """

    stub:
    def args = task.ext.args ?: ''
    """
    mkdir numba_cache_dir
    export NUMBA_CACHE_DIR='./numba_cache_dir'

    touch ${msa.baseName}.fasta
    touch ${msa.baseName}.gff
    touch ${msa.baseName}.vcf
    touch ${msa.baseName}.csv
    touch ${msa.baseName}.phylip
    touch ${msa.baseName}.recombination_predictions.embl
    touch ${msa.baseName}.branch_base_reconstruction.embl
    touch ${msa.baseName}.final_tree.tre
    touch ${msa.baseName}.node_labelled.final_tree.tre

    cat <<-END_VERSIONS > versions.yml
    "${task.process}":
        gubbins: \$(run_gubbins.py --version 2>&1)
    END_VERSIONS
    """
}
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/modules/local/gubbins/

Name	Size	Changed	Rights	Owner
..		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
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meta.yml	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy

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APPLICATION CONT.

```
//
// Perform Snippy analysis by species per clsuter
//

/*
~~~
```

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/workflows/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
poodle.nf	15 KB	9/2/2026 1:19:55 PM	rwxr-xr-x	n.yengalareddy

```
/*
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    VALIDATE INPUTS
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
*/

def validateParameters() {
    // Validate input parameters
    WorkflowPoodle.initialise(params, log)

    // Check input path parameters to see if they exist
    def checkPathParamList = [ params.input ]
    checkPathParamList.each { param -> if (param) { file(param, checkIfExists: true) } }

    // Check mandatory parameters
    if (!params.input) { exit 1, 'Input samplesheet not specified!' }

    def valid_annotation_formats = ['gff', 'split_gff', 'genbank']
    if (!(params.annotation_format in valid_annotation_formats)) {
        exit 1, "ERROR: --annotation_format must be one of: ${valid_annotation_formats.join(', ')}"
    }
}

/*
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    CONFIG FILES
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
*/

/*
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    IMPORT LOCAL MODULES/SUBWORKFLOWS
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
*/

include { CLEAN_TREE as SNIPPY_TREE    } from '../modules/local/cleantree'
include { CLEAN_TREE as GUBBINS_TREE   } from '../modules/local/cleantree'
include { GENEDISTS                    } from '../modules/local/genedists'
include { LINKAGES as LINKAGES_SNIPPY  } from '../modules/local/linkages.nf'
include { LINKAGES as LINKAGES_GUBBINS } from '../modules/local/linkages.nf'
include { CONSTANTSITES                } from '../modules/local/constantsites/main'
include { YAML_BOTH                    } from '../modules/local/report/yaml_both.nf'
include { YAML_GUB_ONLY                } from '../modules/local/report/yaml_gub_only.nf'
include { YAML_MASH_ONLY               } from '../modules/local/report/yaml_mash_only.nf'
include { YAML_NEITHER                 } from '../modules/local/report/yaml_neither.nf'
include { QUARTO_BOTH                  } from '../modules/local/report/quarto_both.nf'
include { QUARTO_GUB_ONLY              } from '../modules/local/report/quarto_gub_only.nf'
include { QUARTO_MASH_ONLY             } from '../modules/local/report/quarto_mash_only.nf'
include { QUARTO_NEITHER               } from '../modules/local/report/quarto_neither.nf'
include { SNPDISTS as SNPDISTS_SNIPPY  } from '../modules/local/snpdists/main'
include { SNPDISTS as SNPDISTS_GUBBINS } from '../modules/local/snpdists/main'
include { IQTREE as IQTREE_SNIPPY      } from '../modules/local/iqtree/main'
include { IQTREE as IQTREE_GUBBINS     } from '../modules/local/iqtree/main'
include { GUBBINS                      } from '../modules/local/gubbins/main'
include { SNPSITES                     } from '../modules/local/snpsites/main'
include { PANAROO_RUN                  } from '../modules/local/panaroo/run/main'
include { MASHTREE                     } from '../modules/local/mashtree/main'

// SUBWORKFLOW: Consisting of a mix of local and nf-core/modules
//
include { INPUT_CHECK                  } from '../subworkflows/local/input_check'
include { SNIPPY_CLUSTERS              } from '../subworkflows/local/snippyclusters.nf'

/*
```

```
workflow POODLE {

    validateParameters()
    ch_input = file(params.input)
    def summary_params = NfcoreSchema.paramsSummaryMap(workflow, params)

    ch_versions = Channel.empty()

    //
    // SUBWORKFLOW: Read in samplesheet, validate and stage input files
    //
    INPUT_CHECK (
        ch_input
    )
    ch_versions = ch_versions.mix(INPUT_CHECK.out.versions)

    // SUBWORKFLOW: Snippy run and Snippy core (verifying prior results and reference)
    //
    SNIPPY_CLUSTERS(
        INPUT_CHECK.out.final_input_files
    )
    ch_versions = SNIPPY_CLUSTERS.out.versions

    //
    // MODULE: Core SNP Distances
    //
    SNPDISTS_SNIPPY(
        SNIPPY_CLUSTERS.out.aln
    )
    ch_versions = ch_versions.mix(SNPDISTS_SNIPPY.out.versions.first())

    //
    // MODULE: Core SNP Linkages
    //
    ch_snp_report_dists = SNPDISTS_SNIPPY.out.tsv
    .join(SNIPPY_CLUSTERS.out.snippy_txt)

    LINKAGES_SNIPPY(
        ch_snp_report_dists
    )
    ch_versions = ch_versions.mix(LINKAGES_SNIPPY.out.versions.first())

    //
    // MODULE: Output count of constant sites (suitable for IQ-TREE -fconst)
    //
    CONSTANTSITES(
        SNIPPY_CLUSTERS.out.clean_full_aln
    )
    ch_versions = ch_versions.mix(CONSTANTSITES.out.versions.first())

    // Convert constant sites file to value
    const_ch = CONSTANTSITES.out.constant_sites.map { meta, p -> tuple(meta, p.text.trim())}
```

```
// MODULE: Gene-presence abscence with Panaroo
//
// Collect annotation files by species and cluster and add assembly if the annotations are split gffs
INPUT_CHECK.out.final_input_files
    .map { meta, reads, assembly, annotation, reference ->

        def panaroo_files
        def panaroo_line

        if (params.annotation_format == 'gff') {
            panaroo_files = [ annotation ]
            panaroo_line  = annotation.getName()
        }
        else if (params.annotation_format == 'split_gff') {
            if (!assembly || assembly.isEmpty()) {
                exit 1, "ERROR: No assembly found for sample '${meta.id}' but --annotation_format=split_gff"
            }

            def assembly_file = assembly instanceof List ? assembly[0] : assembly
            def annotation_file = annotation instanceof List ? annotation[0] : annotation

            panaroo_files = [ annotation_file, assembly_file ]
            panaroo_line  = "${annotation_file.getName()} ${assembly_file.getName()}"
        }
        else if (params.annotation_format == 'genbank') {
            panaroo_files = [ annotation ]
            panaroo_line  = annotation.getName()
        }
        else {
            exit 1, "ERROR: Unsupported annotation_format: ${params.annotation_format}"
        }

        tuple(
            [species: meta.species, cluster_id: meta.cluster_id],
            panaroo_files,
            panaroo_line
        )
    }
    .groupTuple(by: 0)
    .map { meta, annotation_files, input_lines ->
        tuple(meta, annotation_files.flatten(), input_lines)
    }
    .set { ch_panaroo_inputs }
```

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APPLICATION CONT.

```
#!/usr/bin/env nextflow
/*
~~~
```

APPLICATION CONT.

```
// Export these variables to prevent local Python/R libraries from conflicting with those in the container
// The JULIA depot path has been adjusted to a fixed path `/usr/local/share/julia` that needs to be used for packages in the container.
// See https://apeltzer.github.io/post/03-julia-lang-nextflow/ for details on that. Once we have a common agreement on where to keep Julia packages, this is adjustable.

env {
    PYTHONNOUSERSITE = 1
    R_PROFILE_USER   = "/.Rprofile"
    R_ENVIRON_USER   = "/.Renviron"
    JULIA_DEPOT_PATH = "/usr/local/share/julia"
}

// Capture exit codes from upstream processes when piping
process.shell = ['/bin/bash', '-euo', 'pipefail']

timeline {
    enabled = true
    file    = "${params.tracedir}/execution_timeline.html"
}
report {
    enabled = true
    file    = "${params.tracedir}/execution_report.html"
}
trace {
    enabled = true
    file    = "${params.tracedir}/execution_trace.txt"
}
dag {
    enabled = true
    file    = "${params.tracedir}/pipeline_dag.html"
}

manifest {
    name            = 'MDHHS-Bioinformatics/poodle'
    author          = "Karla Vasco, Douglas Maldonado-Torres, Heather Blankenship & Arianna Miles-Jay"
    homePage        = 'https://github.com/MDHHS-Bioinformatics/poodle'
    description     = "Parallel SNP and pangenome analysis of bacterial clusters"
    mainScript      = 'main.nf'
    nextflowVersion = '!>=22.10.1'
    version         = '1.0.0'
    doi             = 'https://doi.org/10.5281/zenodo.20857592'
}

// Load modules.config for DSL2 module specific options
includeConfig 'conf/modules.config'
```

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

Name	Size	Changed	Rights	Owner
..		9/4/2026 11:16:24 AM	rwxr-sr-x	n.yengalareddy
assets		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
bin		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
conf		9/2/2026 1:15:36 PM	rwxr-sr-x	n.yengalareddy
docs		9/2/2026 1:27:14 PM	rwxr-sr-x	n.yengalareddy
input		9/1/2026 2:28:15 PM	rwxr-sr-x	n.yengalareddy
lib		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
modules		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
output		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
quarto_templates		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
work		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
workflows		9/2/2026 1:19:55 PM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATIONS.md	4 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	2 KB	9/2/2026 1:19:11 PM	rw-r--r--	n.yengalareddy
modules.json	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
nextflow.config	7 KB	9/2/2026 1:16:07 PM	rwxr-xr-x	n.yengalareddy
nextflow_schema.json	17 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
poodle.40887634.err	1 KB	9/2/2026 10:50:11 AM	rw-r--r--	n.yengalareddy
poodle.40887634.out	91 KB	9/2/2026 10:58:00 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
README.md	11 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
run_poodle.sh	2 KB	9/1/2026 2:38:36 PM	rw-r--r--	n.yengalareddy
run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

Name	Size	Changed	Rights	Owner
..		9/4/2026 11:16:24 AM	rwxr-sr-x	n.yengalareddy
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conf		9/2/2026 1:15:36 PM	rwxr-sr-x	n.yengalareddy
docs		9/2/2026 1:27:14 PM	rwxr-sr-x	n.yengalareddy
input		9/1/2026 2:28:15 PM	rwxr-sr-x	n.yengalareddy
lib		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
modules		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
output		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
quarto_templates		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
work		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
workflows		9/2/2026 1:19:55 PM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATIONS.md	4 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	2 KB	9/2/2026 1:19:11 PM	rw-r--r--	n.yengalareddy
modules.json	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
nextflow.config	7 KB	9/2/2026 1:16:07 PM	rwxr-xr-x	n.yengalareddy
nextflow_schema.json	17 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
poodle.40887634.err	1 KB	9/2/2026 10:50:11 AM	rw-r--r--	n.yengalareddy
poodle.40887634.out	91 KB	9/2/2026 10:58:00 AM	rw-r--r--	n.yengalareddy
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run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

```
#!/usr/bin/bash

#SBATCH --account=bphl-umbrella
#SBATCH --qos=bphl-umbrella
#SBATCH --job-name=poodle_run
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=40
#SBATCH --mem=200gb
#SBATCH --time=48:00:00
#SBATCH --output=poodle.%j.out
#SBATCH --error=poodle.%j.err
#SBATCH --mail-user=Nikhil.Yengala@flhealth.gov
#SBATCH --mail-type=FAIL,END

set -euo pipefail

module load prokka/1.14.6

cd /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

MANIFEST=input/annotations/prokka_samplesheet.csv

while IFS=$'\t' read -r SAMPLE ASSEMBLY GENUS SPECIES; do
  prokka \
    --outdir input/annotations/${SAMPLE} \
    --prefix ${SAMPLE} \
    --genus ${GENUS} \
    --species ${SPECIES} \
    --cpus ${SLURM_CPUS_PER_TASK} \
    --force \
    ${ASSEMBLY}
done < "$MANIFEST"
```

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APPLICATION CONT.

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subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
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nextflow_schema.json	17 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
poodle.40887634.err	1 KB	9/2/2026 10:50:11 AM	rw-r--r--	n.yengalareddy
poodle.40887634.out	91 KB	9/2/2026 10:58:00 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
README.md	11 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
run_poodle.sh	2 KB	9/1/2026 2:38:36 PM	rw-r--r--	n.yengalareddy
run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

```
#!/usr/bin/bash

#SBATCH --account=bphl-umbrella
#SBATCH --qos=bphl-umbrella
#SBATCH --job-name=poodle_run
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=40
#SBATCH --mem=200gb
#SBATCH --time=48:00:00
#SBATCH --output=poodle.%j.out
#SBATCH --error=poodle.%j.err
#SBATCH --mail-user=Nikhil.Yengala@flhealth.gov
#SBATCH --mail-type=FAIL,END

set -euo pipefail

module load nextflow
module load singularity

export NXF_SINGULARITY_CACHEDIR=/blue/bphl-florida/n.yengalareddy/cache/Nextflow/singularity
mkdir -p "$NXF_SINGULARITY_CACHEDIR"

# POODLE directory
cd /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

# Run POODLE with FASTQ samplesheet
nextflow run main.nf \
  --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/samplesheet.csv \
  --outdir /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output \
  -profile singularity \
  -resume
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/

Name	Size	Changed	Rights	Owner
..		9/4/2026 11:16:24 AM	rwxr-sr-x	n.yengalareddy
assets		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
bin		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
conf		9/2/2026 1:15:36 PM	rwxr-sr-x	n.yengalareddy
docs		9/2/2026 1:27:14 PM	rwxr-sr-x	n.yengalareddy
input		9/1/2026 2:28:15 PM	rwxr-sr-x	n.yengalareddy
lib		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
modules		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
output		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
quarto_templates		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
subworkflows		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
test		8/31/2026 12:58:43 PM	rwxr-sr-x	n.yengalareddy
work		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
workflows		9/2/2026 1:19:55 PM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CITATIONS.md	4 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
main.nf	2 KB	9/2/2026 1:19:11 PM	rw-r--r--	n.yengalareddy
modules.json	3 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
nextflow.config	7 KB	9/2/2026 1:16:07 PM	rwxr-xr-x	n.yengalareddy
nextflow_schema.json	17 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
poodle.40887634.err	1 KB	9/2/2026 10:50:11 AM	rw-r--r--	n.yengalareddy
poodle.40887634.out	91 KB	9/2/2026 10:58:00 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
README.md	11 KB	8/31/2026 12:58:43 PM	rw-r--r--	n.yengalareddy
run_poodle.sh	2 KB	9/1/2026 2:38:36 PM	rw-r--r--	n.yengalareddy
run_prokka.sh	1 KB	9/1/2026 2:46:23 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
Acinetobacter_baumannii		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
pipeline_info		9/2/2026 10:58:00 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy

```
sbatch run_prokka.sh
sbatch run_poodle.sh
```

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
Acinetobacter_baumannii		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
pipeline_info		9/2/2026 10:58:00 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy

Nextflow workflow report

[hungry_sax]

Workflow execution completed successfully!

Run times

02-Sep-2026 13:50:12 - 02-Sep-2026 13:58:00 (duration: **7m 47s**)

62 succeeded

Nextflow command

```
nextflow run main.nf --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/samplesheet.csv --outdir /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output -profile singularity -resume
```

CPU-Hours	1.8
Launch directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026
Work directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/work
Project directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026
Script name	main.nf
Script ID	c43a2cf7436fae342ea32b620b0b6eb3
Workflow name	MDHHS-Bioinformatics/poodle
Workflow version	1.0.0
Workflow description	Parallel SNP and pangenome analysis of bacterial clusters
Workflow author	Karla Vasco, Douglas Maldonado-Torres, Heather Blankenship & Arianna Miles-Jay
Workflow home page	https://github.com/MDHHS-Bioinformatics/poodle
Workflow DOI	https://doi.org/10.5281/zenodo.20857592
Workflow session	c5cd1eda-f84d-4990-8f86-03145ad1d766
Workflow profile	singularity
Nextflow version	version 26.04.3, build 12259 (28-05-2026 08:48 UTC)

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/

Name	Size	Changed	Rights	Owner
..		9/2/2026 1:19:11 PM	rwxr-sr-x	n.yengalareddy
Acinetobacter_baumannii		9/2/2026 10:51:36 AM	rwxr-sr-x	n.yengalareddy
pipeline_info		9/2/2026 10:58:00 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

PoODLE (Phylogenomic Overview for the Detection of Linkages for Epidemiologists)

Whole-Genome Sequencing Report

Repository: https://github.com/MDHHS-Bioinformatics/poodle

PUBLISHED
September 2, 2026

Disclaimer

This report is only for investigation and epidemiology use. It should NOT be used for diagnosis, treatment, or management of patients and should NOT be included in any patient health record.

The data presented in this report has not been validated or subjected to CLIA standards for diagnosing and treating disease. Methods outlined throughout the report are still in development and may be subject to change.

Genomic Assessment

Cluster analysis was performed on 5 samples of the species *Shigella_sonnei* in the cluster named **HC42-C1**. Core genome and pangenome profiles were compared between isolates.

Core-SNP Tree



Figure 1. *Maximum-likelihood phylogeny inferred from core SNPs identified with Snippy-core. The tree was reconstructed in IQ-TREE using the GTR+G4 substitution model with constant sites included. Branch lengths represent substitutions per site across the core genome alignment. Nodes with <50% bootstrap support were collapsed into polytomies. Very short or near-zero branches indicate unresolved relationships among highly similar isolates. Because recombination and horizontal gene transfer may obscure vertical inheritance, branch lengths and fine-scale relationships should be interpreted cautiously. The reference genome was excluded, the tree was midpoint-rooted, and visualization was performed using Phylocanvas.*

Pangenome profile

The Pangenome summary statistics for cluster HC42-C1 are as follows:

```
Core genes  (99% <= strains <= 100%)    4217
Soft core genes (95% <= strains < 99%)  0
Shell genes (15% <= strains < 95%)      467
Cloud genes (0% <= strains < 15%)       0
Total genes (0% <= strains <= 100%) 4684
```



Figure 2. *Gene presence–absence heatmap generated from a pangenome analysis performed with Panaroo. Blue indicates gene presence and white indicates absence. Columns represent genes and rows represent isolates, ordered according to the core SNP phylogeny. Gene names and annotations can be interactively displayed by hovering over individual tiles. Visualization was generated using ggplot2 and plotly.*

Core SNP distance matrix

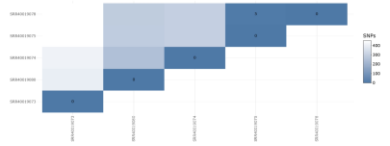


Figure 3. *Heatmap of pairwise core SNP distances among isolates. Values from 0–40 SNPs are displayed. Distances were calculated from the Snippy-core SNP alignment using snp-dists. Rows and columns are ordered according to the core SNP phylogeny. Pairwise SNP differences can be interactively displayed by hovering over individual cells. Visualization was generated using ggplot2 and plotly.*

Gene presence-absence distance matrix

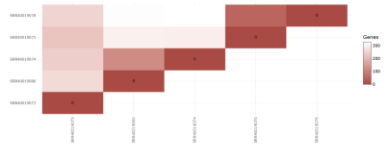


Figure 4. *Heatmap of pairwise gene content distances derived from the Panaroo gene presence–absence matrix. Values from 0–40 gene differences are displayed. Rows and columns are ordered according to the core SNP phylogeny. Exact pairwise gene differences can be interactively displayed by hovering over individual cells. Visualization was generated using ggplot2 and plotly.*

Methods

SNP Analysis

Reference-based SNP analysis was accomplished using Snippy v4.6.0[1] within a species-specific clusters. Maximum-likelihood phylogenies were inferred with IQTREE2 v2.4.0[2] and pairwise SNP distances were calculated using snp-dists v0.8.2[3].

Pangenome Analysis

Gene presence-absence profiles were generated using Panaroo v1.6.0[4] in strict mode, using 95% sequence identity for clustering and a 70% protein family sequence identity threshold.

References

1. https://github.com/tseemann/snippy
2. https://github.com/iqtree/iqtree2
3. https://github.com/tseemann/snp-dists
4. https://github.com/gtonkinhill/panaroo

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

/.../n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/linkages/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1_linkag...	1 KB	9/2/2026 10:57:33 AM	rw-r--r--	n.yengalareddy

sample	species	cluster_id	ref_genome_fraction	ref_alignment_qc	min_dist	strong_linkages	intermediate_linkages	lineage_level
SRR40019073	Shigella_sonnei	HC42-C1	0.9658772152407005	PASS	394	None	None	None
SRR40019074	Shigella_sonnei	HC42-C1	0.9460296844724028	WARN	246	None	None	None
SRR40019075	Shigella_sonnei	HC42-C1	0.9899257120861392	PASS	5	SRR40019078 (5)	None	None
SRR40019078	Shigella_sonnei	HC42-C1	0.9604043758328931	PASS	5	SRR40019075 (5)	None	None
SRR40019080	Shigella_sonnei	HC42-C1	0.9367168387409673	WARN	246	None	None	None

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

/.../n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/panaroo/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1_gene_...	77 KB	9/2/2026 10:51:13 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_gene_...	1 KB	9/2/2026 10:51:13 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_gene_...	668 KB	9/2/2026 10:51:13 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_summ...	1 KB	9/2/2026 10:51:13 AM	rw-r--r--	n.yengalareddy

	SRR40019	SRR40019	SRR40019	SRR40019	SRR40019075
SRR40019080	0	262	121	325	301
SRR40019073	262	0	241	253	225
SRR40019074	121	241	0	328	298
SRR40019078	325	253	328	0	50
SRR40019075	301	225	298	50	0

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

/.../n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/snippy_core/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:57:59 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.aln	5 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.full.aln	28,337 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.iqtree	6 KB	9/2/2026 10:57:49 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.nwk	1 KB	9/2/2026 10:57:49 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.tab	21 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.tre	1 KB	9/2/2026 10:57:51 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.txt	1 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1.vcf	55 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_clean.f...	28,337 KB	9/2/2026 10:57:31 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_consta...	1 KB	9/2/2026 10:57:32 AM	rw-r--r--	n.yengalareddy
Shigella_sonnei_HC42-C1_dist.tsv	1 KB	9/2/2026 10:57:32 AM	rw-r--r--	n.yengalareddy

CHR	POS	REF	SRR40019073	SRR40019074	SRR40019075	SRR40019078	SRR40019080
contig00001	12932	A	A	G	A	A	A
contig00001	19301	T	C	T	T	T	T
contig00001	19991	C	C	C	C	C	T
contig00001	23059	G	A	G	G	G	G
contig00001	23226	G	G	G	G	G	A
contig00001	27039	G	A	A	G	G	A
contig00001	28514	C	C	T	C	C	C
contig00001	36410	C	T	C	C	C	C
contig00001	41947	A	G	A	A	A	A
contig00001	44244	G	G	A	G	G	G
contig00001	50889	G	A	G	G	G	G
contig00001	63303	T	T	T	T	T	A
contig00001	72376	G	G	A	G	G	G
contig00001	76149	C	C	T	C	C	C
contig00002	4890	C	C	T	C	C	C
contig00002	6161	A	G	A	A	A	A
contig00002	41311	C	C	C	C	C	T
contig00002	43919	C	G	C	C	C	C
contig00002	52321	C	T	C	C	C	C
contig00002	55963	G	C	C	G	G	C
contig00002	61136	C	T	C	C	C	C
contig00002	61835	T	C	C	T	T	C
contig00002	65871	T	C	T	T	T	T
contig00003	4754	G	A	A	G	G	A
contig00003	14058	C	T	C	C	C	C
contig00003	16473	T	T	T	T	T	G
contig00003	24218	C	C	C	C	C	T
contig00003	30962	T	A	A	T	T	A
contig00003	33431	T	G	T	T	T	T

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APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
linkages		9/2/2026 10:57:33 AM	rwxr-sr-x	n.yengalareddy
panaroo		9/2/2026 10:51:13 AM	rwxr-sr-x	n.yengalareddy
snippy_core		9/2/2026 10:57:51 AM	rwxr-sr-x	n.yengalareddy
snippy_run		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei_HC42-C1.html	7,672 KB	9/2/2026 10:57:59 AM	rw-r--r--	n.yengalareddy

/.../repos/bphl-molecular/analysis/poodle1026/output/Shigella_sonnei/HC42-C1/snippy_run/SRR40019073/

Name	Size	Changed	Rights	Owner
..		9/2/2026 10:53:43 AM	rwxr-sr-x	n.yengalareddy
SRR40019073.aligned.fa	4,730 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.consensus.fa	4,791 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.consensus.subs.fa	4,790 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.csv	63 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.log	4 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.tab	76 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.txt	1 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy
SRR40019073.vcf	253 KB	9/2/2026 10:53:43 AM	rw-r--r--	n.yengalareddy

```
DateTime        2026-09-02T13:50:16
ReadFiles       /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/fastq/SRR40019073_R1.fastq.gz /blue/bphl-
florida/n.yengalareddy/repos/bphl-molecular/analysis/poodle1026/input/fastq/SRR40019073_R2.fastq.gz
Reference       /blue/bphl-florida/n.yengalareddy/repos/bphl-
molecular/analysis/poodle1026/work/85/d269e907b396e80280e99908eadd5c/Shigella_sonnei_HC42-C1.fna
ReferenceSize   4756763
Software        snippy 4.6.0
Variant-COMPLEX 252
Variant-DEL     34
Variant-INS     30
Variant-MNP     19
Variant-SNP     1343
VariantTotal    1678
```

CHROM	POS	TYPE	REF	ALT	EVIDENCE	FTYPE	STRAND	NT_POS	AA_POS	EFFECT	LOCUS_TAG	GENE	PRODUCT
contig00001	19301	snp	T	C	C:92 T:0								
contig00001	23059	snp	G	A	A:125 G:0								
contig00001	27039	snp	G	A	A:98 G:0								
contig00001	36410	snp	C	T	T:70 C:0								
contig00001	41947	snp	A	G	G:96 A:0								
contig00001	50889	snp	G	A	A:108 G:0								
contig00002	6161	snp	A	G	G:89 A:0								
contig00002	43919	snp	C	G	G:60 C:0								
contig00002	52321	snp	C	T	T:88 C:0								
contig00002	55963	snp	G	C	C:104 G:0								
contig00002	61136	snp	C	T	T:104 C:1								
contig00002	61835	snp	T	C	C:117 T:0								
contig00002	65871	snp	T	C	C:94 T:0								
contig00003	4754	snp	G	A	A:106 G:0								
contig00003	14058	snp	C	T	T:108 C:0								
contig00003	30962	snp	T	A	A:112 T:0								
contig00003	33431	snp	T	G	G:80 T:0								
contig00003	42163	snp	G	T	T:61 G:1								

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CONCLUSION

- Fundamentals of Poodle
- Installation and setup of Poodle in HPG
- Successfully executed job query for Poodle
- Generated output files



Advanced Molecular Detection

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Questions?

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